MULTI-MODAL EDGE-AI SENSOR FUSION ARCHITECTURE

Technical Specification & Analytics Deliverables (v4.0)

1. Executive Summary & System Philosophy

Modern cricket training demands high-precision kinematic and kinetic analytics, yet existing solutions are polarized: elite optical systems are prohibitively expensive ($10,000 to $100,000+ USD) and dependent on external cloud infrastructure, while consumer wearable sensors alter natural player mechanics and introduce observer bias. This specification outlines an advanced, non-intrusive, multi-modal tracking architecture designed specifically for indoor practice facilities. By fusing ***Piezoelectric***, ***Infrared (IR)***, ***Ultrasonic***, ***mmWave Radar***, and a 6-unit ***Spatial AI*** vision array via a localized Edge-AI processing pipeline, the system delivers lab-grade telemetry with zero cloud latency and complete immunity to indoor lighting glare.

2. Comprehensive Sensor Matrix & Functional Roles

Each sensor modality addresses distinct physical domains, eliminating individual blind spots through overlapping data collection:

A. Piezoelectric Sensors (Tactile & Impact Dynamics)

Placement: Embedded beneath synthetic turf pitch mats, inside the base of stumps, and structurally bonded to training frames.
Function: Registers physical vibration, mechanical stress, and impact shockwaves. It provides the microsecond-precise master timestamp synchronization for bat-ball impacts and foot-crease landings.

B. Near-Infrared (NIR) Arrays & Light Curtains (Spatial Gating)

Placement: Arrayed across the bowling release line, popping crease, and hitting zone, paired with narrow-bandpass optical filters.
Function: Captures high-speed velocity entry/exit timing and spatial gating coordinates completely unaffected by visual lighting variations.

C. Ultrasonic Proximity Sensors (Spatial Depth & Stance Mapping)

Placement: Mounted on overhead frames and side-nets surrounding the crease.
Function: Emits high-frequency sound waves to execute non-contact distance measurements, continuously tracking player stance positioning and body weight distribution sway.

D. mmWave Radar Modules (Continuous Volumetric Tracking)

Placement: Positioned at bowler-end and keeper-end enclosures utilizing FMCW technology.
Function: Tracks continuous 3D flight paths from bowler release through pitch bounce to the bat, extracting micro-Doppler frequency modulations for spin and velocity profiling.

E. Spatial AI Units (6x OAK-D Lite Multi-Camera Array)

Placement: Distributed perimeter network (2 diagonal facing crease, 2 down pitch/bowler end, 1 overhead, 1 wide side-on).
Function: Executes localized neural inference and binocular depth disparity mapping to eliminate body occlusion, provide full 3D skeletal pose estimation, and supply high-resolution visual confirmation of bat-face orientation.

3. Integrated Analytics Suite

3.1 Bowling Analytics

- **Ball Speed Profile:** Release speed, speed through air, post-bounce deceleration.
- **3D Ball Trajectory:** Line and length, pitch-map zones, beehive plots, wagon wheels for where the ball finishes.
- **Swing and Seam Movement Indicators:** Lateral deviation before and after bounce, plus spin/swerve trends from mmWave micro-Doppler and stereo depth.
- **Release Point and Run-Up Metrics:** Release height, front-foot landing position, approach speed, over/around-the-wicket, wide-on-crease events.
- **Bowling Action Biomechanics:** Arm angle, shoulder-hip rotation, alignment, head position stability, and whether the action is repeatable or drifting over a spell.
- **Accuracy and Consistency KPIs:** Percentage hitting target zones, variability of length/line, spell-wise heat maps and trend charts.

Coach Illustration: "For every over, you'll see exactly where each ball landed, how much it moved, at what speed, and how your action changed ball-by-ball."

3.2 Batting Analytics

On the batting side, fusing bat-tracking, pose estimation, impact timing and ball data lets you deliver, per shot:

- **Shot Type Classification:** Drive, pull, cut, sweep, etc., and outcome tagging (beaten, edge, controlled, lofted, boundary intent).
- **Bat Swing Path and Speed:** Downswing plane, back-lift angle, bat face orientation at impact, follow-through direction.
- **Contact Quality:** Impact location on bat (sweet spot vs toe/edge), timing vs ideal contact window, bat-ball distance at closest approach.
- **Footwork and Balance:** Initial stance, front/back-foot movement, distance to pitch of the ball, head position over base of support, balance drift during stroke.
- **Scoring Zones and Beehive:** Where runs are scored for different lengths and lines, preferred scoring areas, and where the batter struggles.
- **Decision-Making Indicators:** Balls attacked vs left, false shots percentage, repeatability of shot selection for specific line/length combinations.

4. Edge-AI Data Fusion Pipeline Architecture

Raw data streams arrive asynchronously from diverse physical mediums. The local Edge-AI processing box executes a three-stage sequential fusion pipeline:

Pipeline Stage	Data Processing Logic	Operational Output
Stage A: Low-Level Time Synchronization	Hardware timestamping binds all incoming sensor ticks to a unified clock. A Piezoelectric impact spike acts as a master trigger, isolating a tight 50ms temporal window across all 6 spatial cameras and radar streams.	Eliminates drift across optical, acoustic, tactile, and radar data streams.
Stage B: Mid-Level Cross-Validation	The Edge-AI cross-references tactile vibration against IR light curtains, mmWave radar, and Spatial AI depth maps. If a vibration occurs without an intersecting ball trajectory, it is classified as a ground tap.	100% elimination of false positives and spurious data logging.
Stage C: High-Level Machine Learning	Lightweight neural networks evaluate fused feature vectors from the 6-camera array and radar to categorize stroke execution, delivery classification, sweet-spot accuracy, and kinetic efficiency.	Real-time tactical metrics pushed instantly to local coach tablets (<20ms latency).

5. Comparative Analysis with Market Alternatives

System Category	Core Functions	Estimated Cost	Accuracy & Limitations
High-End Optical Labs (e.g., Hawk-Eye)	Full 3D spatial ball tracking and biomechanical skeletal pose analysis.	Extremely High ($10k - $100k+ USD)	Sub-millimeter spatial accuracy; highly vulnerable to indoor lighting glare and expensive calibration overhead.
Wearable Smart-Bat Sensors (e.g., Str8bat)	Bat speed, downswing plane, twist angles, and impact sweet-spot index.	Consumer-Tier ($60 - $80 USD)	High internal movement fidelity; completely blind to ball flight, pitch behavior, and bowler metrics.
Pitch-Embedded Turf Arrays (e.g., PitchVision)	Pitch coordinates, deviation, seam movement, and impact match stats.	Moderate to High (Commercial Grade)	High impact accuracy; requires permanent, disruptive structural civil modification of the practice pitch.
Proposed Multi-Modal Edge-AI Concept	Non-intrusive, multi-modal tracking: piezo impacts, NIR gating, ultrasonic stance, mmWave radar, and 6-camera OAK-D spatial AI.	Moderate (~$1,650 - $1,900 USD)	Exceptional volumetric & temporal accuracy, zero wearables, complete immunity to lighting glare, and zero cloud latency.

6. Facility Implementation & Environmental Resilience

Indoor facilities frequently suffer from harsh spotlight glare, thermal fluctuations, and heavy visual shadows. The proposed architecture overcomes these limitations through structural engineering:

- **Optical Immunity:** Narrow-bandpass filters on NIR and Spatial AI sensors block stray visible light and thermal glare from overhead focus spotlights.
- **Acoustic & Mechanical Independence:** Because Piezoelectric and mmWave radar operate independently of ambient light, tracking integrity remains absolute under any indoor lighting condition.
- **Multi-Camera Redundancy:** The 6-unit Spatial AI array ensures that even if a batsman's body momentarily occludes one angle, overlapping disparity maps maintain uninterrupted 3D tracking.
- **Edge Autonomy:** Local processing ensures complete data privacy and instantaneous feedback without reliance on high-bandwidth internet connectivity.